

Session 6

Aim

To study process states, process termination, zombie and orphan processes, and non-preemptive CPU scheduling.

```
bhavan@LAPTOP-FGDR116:~$ ps
  PID TTY          TIME CMD
 334 pts/0    00:00:00 bash
 453 pts/0    00:00:00 ps

bhavan@LAPTOP-FGDR116:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1        0  0 03:30 ?        00:00:01 /sbin/init
root           2        1  0 03:30 ?        00:00:00 /init
root           6        1  0 03:30 ?        00:00:00 /usr/lib/systemd/systemd-journald
root          51        1  1 03:30 ?        00:00:00 /usr/lib/systemd/systemd-udev
root         187        1  1 03:30 ?        00:00:00 /usr/lib/systemd/systemd-resolved
systemd+     126        1  0 03:30 ?        00:00:00 /usr/lib/systemd/systemd-timesyncd
systemd+     127        1  0 03:30 ?        00:00:00 /usr/sbin/cron -f -P
message+    188        1  0 03:30 ?        00:00:00 dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
root        198        1  1 03:30 ?        00:00:00 /usr/lib/systemd/systemd-logind
root        281        1  1 03:30 ?        00:00:00 /usr/libexec/ssl-proc-service
root        218        1  0 03:30 ?        00:00:00 /sbin/agetty -a -p -- \u --noclear - linux
syslog      220        1  0 03:30 ?        00:00:00 /usr/sbin/rsyslogd -n -iNONE
root        230        1  2 03:30 ?        00:00:00 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root        280        1  1 03:31 ?        00:00:00 /usr/lib/systemd/systemd-timedated
root        331        2  0 03:31 ?        00:00:00 /init
root        332        331  0 03:31 ?        00:00:00 /init
bhavan      334        332  0 03:31 pts/0    00:00:00 -bash
root        335        2  0 03:31 pts/1    00:00:00 /bin/login -f
bhavan      384        1  1 03:31 ?        00:00:00 /usr/lib/systemd/systemd --user
bhavan      381        384  0 03:31 ?        00:00:00 (sd-pam)
bhavan      411        335  0 03:31 pts/1    00:00:00 -bash
bhavan      454        334  0 03:31 pts/0    00:00:00 ps -ef

bhavan@LAPTOP-FGDR116:~$ top
top - 03:31:42 up 0 min, 1 user, load average: 0.61, 0.38, 0.06
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
Cpu(s):  0.0 us,  0.1 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
Mem: 7754.6 total, 6324.8 free, 503.7 used, 1067.0 buff/cache
Mem Swap: 2048.0 total, 2048.0 free,  0.0 used, 7250.0 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM     time+  COMMAND
    1 root        20   0 21888 13172 9700  S   0.0   0.2   0:01.60  systemd
    2 root        20   0 3180  2212 2072  S   0.0   0.0   0:00.03  init-systemd(Ub
    6 root        20   0 3196  2124 2000  S   0.0   0.0   0:00.00  init
   61 root       19  -1 42188 16120 15068  S   0.0   0.2   0:00.29  systemd-journal
  187 root        20   0 25152 6536 5166  S   0.0   0.1   0:00.37  systemd-udev
  126 systemd+    20   0 21344 13196 11068  S   0.0   0.2   0:00.25  systemd-resolve
  127 systemd+    20   0 91836 7920 6952  S   0.0   0.1   0:00.13  systemd-timesyn
  187 root        20   0 4200  2684 2432  S   0.0   0.0   0:00.01  cron
  188 message+    20   0 3644  5400 4700  S   0.0   0.1   0:00.12  dbus-daemon
  198 root        20   0 17988 8846 7800  S   0.0   0.1   0:00.22  systemd-logind
  281 root        20   0 1756372 12684 11040  S   0.0   0.2   0:00.25  ssl-proxy-service
  218 root        20   0 3124  1972 1836  S   0.0   0.0   0:00.01  atetty
  220 syslog      20   0 222516 5072 4580  S   0.0   0.1   0:00.19  rsyslogd
  230 root        20   0 187836 23172 13716  S   0.0   0.3   0:00.43  unattended-upgr
  331 root        20   0 3184  1116  980  S   0.0   0.0   0:00.00  SessionLeader
  332 root        20   0 1200  1252 1100  S   0.0   0.0   0:00.01  Relay(130)
  334 teja        20   0 6876  5388 3668  S   0.0   0.1   0:00.07  bash
  335 root        20   0 6704  4604 3820  S   0.0   0.1   0:00.03  login
  384 teja        20   0 20328 11564 9448  S   0.0   0.1   0:00.25  systemd
  381 teja        20   0 21160 3572 1000  S   0.0   0.0   0:00.00  (sd-pam)
  411 teja        20   0 4800  5376 3708  S   0.0   0.1   0:00.07  bash
  454 teja        20   0 9300  5602 3500  S   0.0   0.1   0:00.00  top
```

```
top - 03:34:00 up 3 min, 1 user, load average: 0.26, 0.17, 0.07
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie
Cpu(s):  0.0 us,  0.1 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
Mem: 7754.6 total, 6200.6 free, 543.8 used, 1067.1 buff/cache
Mem Swap: 2048.0 total, 2048.0 free,  0.0 used, 7210.0 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM     time+  COMMAND
    1 root        20   0 21888 13172 9700  S   0.0   0.2   0:01.60  systemd
    2 root        20   0 3180  2212 2072  S   0.0   0.0   0:00.03  init-systemd(Ub
    6 root        20   0 3196  2124 2000  S   0.0   0.0   0:00.00  init
   61 root       19  -1 42188 16120 15068  S   0.0   0.2   0:00.29  systemd-journal
  187 root        20   0 25152 6536 5166  S   0.0   0.1   0:00.37  systemd-udev
  126 systemd+    20   0 21344 13196 11068  S   0.0   0.2   0:00.25  systemd-resolve
  127 systemd+    20   0 91836 7920 6952  S   0.0   0.1   0:00.13  systemd-timesyn
  187 root        20   0 4200  2684 2432  S   0.0   0.0   0:00.01  cron
  188 message+    20   0 3644  5400 4700  S   0.0   0.1   0:00.12  dbus-daemon
  198 root        20   0 17988 8846 7800  S   0.0   0.1   0:00.22  systemd-logind
  281 root        20   0 1756372 12684 11040  S   0.0   0.2   0:00.25  ssl-proxy-service
  218 root        20   0 3124  1972 1836  S   0.0   0.0   0:00.01  atetty
  220 syslog      20   0 222516 5072 4580  S   0.0   0.1   0:00.19  rsyslogd
  230 root        20   0 187836 23172 13716  S   0.0   0.3   0:00.43  unattended-upgr
  331 root        20   0 3184  1116  980  S   0.0   0.0   0:00.00  SessionLeader
  332 root        20   0 1200  1252 1100  S   0.0   0.0   0:00.01  Relay(130)
  334 teja        20   0 6876  5388 3668  S   0.0   0.1   0:00.07  bash
  335 root        20   0 6704  4604 3820  S   0.0   0.1   0:00.03  login
  384 teja        20   0 20328 11564 9448  S   0.0   0.1   0:00.25  systemd
  381 teja        20   0 21160 3572 1000  S   0.0   0.0   0:00.00  (sd-pam)
  411 teja        20   0 4800  5376 3708  S   0.0   0.1   0:00.07  bash
  454 teja        20   0 9300  5602 3500  S   0.0   0.1   0:00.00  top
```

```

GNU nano 2.7.2 zombie.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main()
{
    pid_t pid = fork();
    if (pid > 0)
    {
        printf("Parent process is running\n");
        sleep(20);
    }
    else if (pid == 0)
    {
        printf("Child process completed\n");
        exit(0);
    }
    else
    {
        printf("Fork failed\n");
    }
    return 0;
}

bhavan@LAPTOP-FGDPB110:~$ sleep 20
bhavan@LAPTOP-FGDPB110:~$ sleep 5
bhavan@LAPTOP-FGDPB110:~$ sleep 100
^C
bhavan@LAPTOP-FGDPB110:~$ sleep 100 &
[1] 485
bhavan@LAPTOP-FGDPB110:~$ jobs
[1]+  Running                  sleep 100 &
bhavan@LAPTOP-FGDPB110:~$ nano zombie.c
bhavan@LAPTOP-FGDPB110:~$ gcc zombie.c -o zombie
[1]+  Done                      sleep 100
bhavan@LAPTOP-FGDPB110:~$ ./zombie &
[1] 498
bhavan@LAPTOP-FGDPB110:~$ Child process completed
Parent process is running
bhavan@LAPTOP-FGDPB110:~$ ps -el | grep Z
F S  UID      PID     PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
1 Z   1000      499      498  0  80   0 -    0 -          pts/0        00:00:00 zombie
bhavan@LAPTOP-FGDPB110:~$ |

```

```

bhavan@LAPTOP-FGDPB110:~$ gcc zombie.c -o zombie
[1]+  Done                      sleep 100
bhavan@LAPTOP-FGDPB110:~$ ./zombie &
[1] 498
bhavan@LAPTOP-FGDPB110:~$ Child process completed
Parent process is running
bhavan@LAPTOP-FGDPB110:~$ ps -el | grep Z
F S  UID      PID     PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
1 Z   1000      499      498  0  80   0 -    0 -          pts/0        00:00:00 zombie
bhavan@LAPTOP-FGDPB110:~$ nano orphan.c
bhavan@LAPTOP-FGDPB110:~$ gcc orphan.c -o orphan
bhavan@LAPTOP-FGDPB110:~$ ./orphan
Parent process completed
Child process is running
New parent ID: 332
bhavan@LAPTOP-FGDPB110:~$ |

```

Short theory

A zombie process is a child process that has completed its work, but its parent has not collected its exit status. It remains temporarily in the process table

```
GNU nano 2.9.2 orphan.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main()
{
    pid_t pid = fork();
    if (pid > 0)
    {
        printf("Parent process completed\n");
        exit(0);
    }
    else if (pid == 0)
    {
        sleep(5);
        printf("Child process is running\n");
        printf("New parent ID: %d\n", getppid());
    }
    else
    {
        printf("Fork failed\n");
    }
    return 0;
}
```

```
bhavan@LAPTOP-FGDPB110:~$ gcc zombie.c -o zombie
[1]+  Done                  sleep 100
bhavan@LAPTOP-FGDPB110:~$ ./zombie &
[1] 498
bhavan@LAPTOP-FGDPB110:~$ Child process completed
Parent process is running
bhavan@LAPTOP-FGDPB110:~$ ps -el | grep Z
F S    UID     PID     PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
1 Z    1000     499       498  0  80   0 -    0 -          pts/0        00:00:00 zombie
bhavan@LAPTOP-FGDPB110:~$ nano orphan.c
bhavan@LAPTOP-FGDPB110:~$ gcc orphan.c -o orphan
bhavan@LAPTOP-FGDPB110:~$ ./orphan
Parent process completed
Child process is running
New parent ID: 332
bhavan@LAPTOP-FGDPB110:~$ |
```

```

bhavan@LAPTOP-FGDPB110:~$ gcc orphan.c -o orphan
bhavan@LAPTOP-FGDPB110:~$ ./orphan
Parent process completed
bhavan@LAPTOP-FGDPB110:~$ Child process is running
New parent ID: 332
bhavan@LAPTOP-FGDPB110:~$ ps -eo pid,comm,ni,pri
  PID COMMAND          NI PRI
    1 systemd            0  19
    2 init-systemd(Ub    0  19
    6 init               0  19
   61 systemd-journal   -1  20
  107 systemd-udev      0  19
  126 systemd-resolve    0  19
  127 systemd-timesyn    0  19
  187 cron               0  19
  188 dbus-daemon         0  19
  198 systemd-logind      0  19
  218 agetty              0  19
  220 rsyslogd             0  19
  230 unattended-upgr     0  19
  331 SessionLeader       0  19
  332 Relay(334)          0  19
  334 bash                0  19
  335 login               0  19
  384 systemd             0  19
  385 (sd-pam)            0  19
  411 bash                0  19
  541 ps                  0  19
bhavan@LAPTOP-FGDPB110:~$ nice -n 10 sleep 30 &
[1] 542
bhavan@LAPTOP-FGDPB110:~$ nice -n 10 sleep 30 &
[2] 543
bhavan@LAPTOP-FGDPB110:~$ ps -eo pid,comm,ni,pri | grep sleep
  542 sleep             10  9
  543 sleep             10  9
bhavan@LAPTOP-FGDPB110:~$

```

Session 6 short theory

In this session, we studied process states and process termination. We observed waiting processes using sleep, normal and abnormal termination, zombie processes and orphan processes. We also observed process priority. CPU scheduling helps the operating system select the next process for execution. FCFS executes processes in arrival order, while SJF selects the process with the shortest burst time.